

Validation targets for tropical upwelling and polar vortex strength

Derived from the CESM2.1.5-WACCM6 fixed-SST training data.
Targets anchored on 1996-2014 and verified out-of-sample on 1970-1995.
Thresholds given for both 10-year and 20-year rollouts.

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Summary

Two diagnostics are targeted: the strength of the tropical upwelling branch of the Brewer-Dobson circulation, and the strength of the winter polar vortex. Both are supporting quantities for AIDE-SAI: upwelling sets how fast injected aerosol is lofted and spread, and the vortex sets the high-latitude transport barrier and the descent that removes it.

Every threshold below is computed from the CESM output rather than assumed. The anchor is always half of CESM's own interannual variability, floored at what a rollout of a given length can actually resolve.

The five results that change how the targets should be written

- VTHzm on the CESM h6 tape is ALREADY the eddy heat flux $v'\theta'$. Subtracting $Vzm \cdot THzm$ - the natural reading of its long_name - makes tropical w^* come out downward, which is unphysical. Verified against the raw 3-D daily fields: 15% median error taking it as-is, 1087% if subtracted.
- The two standard estimators of w^* disagree by 10.8% on identical data, and the coarse-grid estimator error is comparable to the target size. The upwelling targets are therefore only meaningful when model and truth go through the same reduction on the same grid.
- CESM's Brewer-Dobson circulation accelerates by 2.1%/decade under prescribed GHG forcing. This inflates the raw interannual sigma by 54%, and it means an absolute upwelling target does not transfer between epochs - CESM 1970-1995 fails a target set on CESM 1996-2014 by 3.5x.
- The polar vortex targets do transfer: all four vortex and polar-cap bias targets pass out of sample at 0.14-0.84x their threshold, and they survive coarse-grid diagnosis with an order of magnitude to spare.
- Of the two mechanism tests, thermal wind (R2) is stable across periods and resolvable in 3 winters; wave driving (R1) is not stable and must be tested by CI overlap rather than containment.

At 10 years the rule is $target(n) = \max(0.5\sigma, 1.96\sigma/\sqrt{n})$. Bias targets loosen by 24% relative to 20 years. The interannual variance ratio becomes untestable and should be replaced by the daily variance ratio; SSW frequency should be stated as an event count with a Poisson interval rather than a rate $\pm 50\%$.

1.1 Tropical upwelling: what is being measured

Air enters the stratosphere through the tropical tropopause, rises slowly, drifts poleward, and descends at middle and high latitudes. That overturning is the Brewer-Dobson circulation (BDC). Its ascending branch is what carries injected sulfate upward and determines how long it stays aloft, so its speed is the single most consequential transport number for AIDE-SAI.

Why not just use the vertical wind?

The zonal-mean vertical velocity $\bar{w}$ is not the velocity that transports tracers. In the stratosphere, large eddy fluxes drive a mean-flow response that almost cancels the Eulerian circulation. The transport-relevant quantity is the RESIDUAL velocity of the Transformed Eulerian Mean (TEM) framework, $\bar{w}^*$, which adds back the eddy contribution (Andrews, Holton & Leovy 1987, eq. 3.5.1):

$$\psi = v' \theta' / (\partial \bar{\theta} / \partial z)$$

$$\bar{w}^* = \bar{w} + (1 / (a \cos \phi)) \cdot \partial / \partial \phi [\cos \phi \cdot \psi]$$

with a the Earth radius, ϕ latitude, and $z = -H \ln(p/p_0)$ the log-pressure height ($H = 7$ km). In this CESM output the eddy term is not a small correction: at 70 hPa in the deep tropics the Eulerian $\bar{w}$ is 0.122 mm/s and the residual $\bar{w}^*$ is 0.196 mm/s, so the eddies supply 38% of the answer.

The primary metric: upward mass flux, not velocity

A velocity at one latitude band is noisy and depends on where you draw the band. The standard BDC metric (Rosenlof 1995; Butchart et al. 2010; the CCMVal protocol) integrates the upward mass transport across a pressure surface over the whole region where the residual flow is upward:

$$M(p) = 2\pi a^2 \int \rho \bar{w}^* \cos \phi \, d\phi \quad \text{over all } \phi \text{ where } \bar{w}^* > 0$$

The integration limits are the 'turnaround latitudes' where $\bar{w}^*$ changes sign - diagnosed here at 35.6°S to 34.5°N at 70 hPa. The result is a mass transport in kg/s, about 9.7×10^9 kg/s, i.e. roughly ten million tonnes of air crossing 70 hPa upward every second.

In pressure coordinates the mass flux per unit area is exactly $-\omega/g$. Because W_{zm} is the log-pressure velocity $w = -H\omega/p$ and the log-pressure density is $\rho_0 = p/(gH)$, the product $\rho_0 \bar{w}^*$ equals $-\bar{\omega}^*/g$ identically. The formulation is therefore exact, not an approximation with H in it.

Mass flux is preferred over $\bar{w}^*$ as the gated metric because it is an integral and so less noisy: its interannual sigma is 2.3% of the mean against 6.5% for the 10°S-10°N $\bar{w}^*$, which makes it the more discriminating test. $\bar{w}^*$ is still reported, because it is the quantity most papers quote.

1.2 Where the numbers come from: CAM history tapes

CAM writes its output to numbered history streams, called TAPES - a term inherited from magnetic tape and still used throughout CESM. Each tape is configured independently: it has its own variable list, its own output frequency, and its own averaging rule. They are named h0, h1, h2 ... and a single run typically writes several, because different diagnostics need different trade-offs between how many variables, how often, and how big.

This run writes four tapes. Only one of them is used for the targets:

Tape	Frequency	Content	Vars	Size (1996-2014)
h0	monthly mean	3-D and surface fields, the general-purpose archive	223	213 GB
h1	hourly mean	3-D dynamics and chemistry on all 70 levels	71	59 TB
h6	daily mean	ZONAL-MEAN TEM diagnostics — used here	16	2 GB
h7	daily mean	3-D subset	35	1.6 TB

The h6 tape is CAM's purpose-built TEM output. It carries Uzm, Vzm, Wzm, THzm, VTHzm, UVzm and UWzm — precisely the input set the standard Transformed Eulerian Mean calculation requires — already reduced to zonal means on the (time, ilev, lat) grid, with the eddy fluxes accumulated inside the model at every 1800 s timestep.

'hist files' versus what is actually on disk

Raw CAM history files are written one per time chunk with every variable of that tape inside, named `case.cam.h6.1996-01-01-00000.nc`, and normally live in a `hist/` directory. Standard CESM post-processing then transposes them into TIME SERIES form: one variable per file spanning the whole period, `case.cam.h6.VTHzm.19960101-20141231.nc`. It is the same data, losslessly reorganised, and it is far more efficient when you want one variable across many years rather than all variables for one month.

This archive contains only the time-series form. There is no `hist/` directory in the run at all — the tree is `archive/atm/proc/tseries/{month_1, day_1, hour_1}`. So 'the hist files' and 'the h6 tape' are not alternatives here: the `tseries` files ARE the history output, and h6 is one of the four tapes within it.

1.3 The fields read, and how they combine

Variable	Symbol	Units	Role in the calculation
Vzm	$\bar{v}$	m/s	zonal-mean meridional wind; gives $\bar{v}^*$ and the continuity cross-check
Wzm	$\bar{w}$	m/s	zonal-mean LOG-PRESSURE vertical velocity, $w = -H\omega/p$; first term of $\bar{w}^*$
THzm	θ	K	zonal-mean potential temperature; supplies $\partial\theta/\partial z$, and polar cap T via $T = \theta(p/p_0)^\kappa$
VTHzm	$v'\theta'$	K m/s	zonal-mean EDDY meridional heat flux; the numerator of ψ , and the R1 predictor
Uzm	$\bar{u}$	m/s	zonal-mean zonal wind; the vortex index and the SSW detector
ilev, hyai, hybi, P0	p	hPa / Pa	the 71 hybrid interface pressures; above ~ 182 hPa $hybi = 0$, so p is exactly $hyai \cdot P0$ and independent of surface pressure
lat	ϕ	deg	192 latitudes on the native f09 grid

Derived quantities

```

z          = -H ln(p/p0)          log-pressure height, H = 7000 m
dthdz      = d(THzm)/dz          centred difference on the interface grid
psi         = VTHzm / dthdz       [m^2/s]
wstar      = Wzm + (1/(a cos.phi)) d/d.phi (cos.phi psi)      [m/s]
w70        = wstar interpolated to 70 hPa (linear in log p), cos-weighted 10S-10N
massflux   = 2 pi a^2 SUM rho0 * max(wstar,0) * cos.phi * d.phi, rho0 = p/(g H)
u60        = Uzm interpolated to 10 hPa and to exactly 60 deg latitude
Tcap       = THzm * (p/p0)^kappa, log-p weighted 10-50 hPa, cos-weighted 60-90 deg

```

Two grid details matter. The TEM fields live on INTERFACE levels (ilev, 71 of them), not the 70 midpoints, so they must not be mixed with lev-based fields without interpolation. And neither 70 hPa nor 10 hPa is a model interface: 70 hPa lies between 66.80 and 80.70 hPa, 10 hPa between 9.55 and 11.86 hPa, so both are interpolated linearly in log p.

Below-surface points are written as the sentinel 1e35 with NO_FillValue attribute, so xarray will not mask them and an unguarded mean returns $\sim 1e33$. 1.34% of cells are affected, all at $p \geq 652$ hPa. Every load in this suite masks with $\text{abs}(x) < 1e20$ before any reduction.

1.4 Why the h6 tape rather than the 3-D fields

The h1 and h7 tapes carry V and T on all 70 levels, so in principle the eddy heat flux could be rebuilt directly and the TEM calculation done from scratch. Four reasons not to, in decreasing order of force.

1. Cost

Rebuilding TEM needs U, V, T and OMEGA at high time resolution. On the hourly tape those four total 7.0 TB for 1996–2014 alone, ~15 TB for the full 45 years; the seven h6 fields total 2.0 GB — a factor of about 3500. The whole 45-year reference here reads ~2 GB and runs in a minute.

2. The model's own accumulation is the better reference

VTHzm is accumulated inside CAM every 1800 s. The 3-D tapes are themselves TIME AVERAGES (h1 hourly, h7 daily, both cell_methods = 'time: mean'), so a rebuild loses covariance inside the averaging window.

3. But how much does that actually matter? — measured

Script 01d computes the same 45–75°N eddy heat flux three ways, on four sample days at four levels, with the levels matched exactly:

Route	Median error vs h6	Median ratio to h6	Reading
Rebuilt from 24 hourly means	2.4 %	0.996	essentially unbiased — hourly resolves the eddy covariance
Rebuilt from the daily mean	2.9 %	0.971	biased ~3% LOW — sub-daily covariance is lost, as expected
Rebuilt WITHOUT matching levels	8–10 %	—	the dominant error is the interface-vs-midpoint offset, not time averaging

Weaker than I first assumed, and worth saying plainly: a careful rebuild reproduces the model's eddy flux to 2–3%. Time averaging is NOT the dominant error — comparing an interface field against a midpoint one is, and that alone costs 8–10%.

4. Two pitfalls a rebuild must handle, both found the hard way

- TIME ALIGNMENT. Both tapes stamp at the END of the interval and carry a degenerate first record (time_bnds = [0,0]). Day d on h6/h7 corresponds to h1 indices d·24–23 ... d·24. The naive d·24 ... d·24+23 is wrong by a full day: with the correct offset the 24-hour mean matches the h7 daily mean to 5×10^{−8}, without it the apparent eddy-flux error inflates from 2.4% to 32%.
- LEVEL GRIDS. The TEM fields are on the 71 INTERFACES (ilev), the 3-D fields on the 70 MIDPOINTS (lev). 'Nearest level' compares 73.75 hPa against 66.80 hPa — a 10% pressure difference across a sharp vertical gradient.

Both routes are used here: h6 for the reference values, the 3-D snapshot route in section 8 — an emulator can be scored no other

1.5 Polar vortex, polar cap temperature and sudden warmings

Vortex strength

The stratospheric polar vortex is the winter westerly jet encircling the pole. Its conventional index is the zonal-mean zonal wind at 60° latitude and 10 hPa - the level and latitude used by the WMO and by essentially all of the SSW literature, which is why they are used unchanged here. Built from Uzm, interpolated to exactly 60° in latitude and to 10 hPa linearly in log p (between interfaces 9.55 and 11.86 hPa).

The southern vortex is reported sign-flipped so that 'westerly' is positive in both hemispheres. It is 3.3× stronger and 2.1× less variable than the northern one, which makes it the sharpest available detector of slow drift.

Polar cap temperature

The mass-weighted mean temperature over 60-90° and 10-50 hPa. It is not read directly: the h6 tape carries potential temperature, so temperature is recovered as $T = \theta (p/p_0)^{\kappa}$ with $\kappa = R/c_p = 0.2858$, then averaged with log-pressure weights across the layer and $\cos \phi$ weights across the cap.

Vortex strength and polar cap temperature are not independent - they are the same bias seen twice, linked by thermal wind. That link is quantified in section 7 and used as a consistency test.

Sudden stratospheric warmings

A major SSW is detected with the Charlton & Polvani (2007) criteria: the daily zonal-mean zonal wind at 60°N/10 hPa reverses to easterly during the extended winter (1 Nov - 31 Mar); consecutive events must be separated by at least 20 consecutive westerly days; and final warmings are excluded by requiring the wind to return westerly for at least 10 consecutive days before 30 April.

SSWs matter here because no RMSE-type metric can express them. A model can have the right mean vortex and the right variance and still never break the vortex down, which would misrepresent both transport and the aerosol residence time in the polar stratosphere.

Eddy heat flux at 100 hPa

The 45-75°N average of VTHzm at 100 hPa measures upward wave activity entering the stratosphere - the forcing that drives both the BDC and vortex breakdown. It carries no target of its own; it is the predictor in the R1 mechanism regression.

2 Data provenance, and four conventions that had to be verified

Two of the four are undocumented in the files and both change the answer qualitatively. They are recorded here because anyone re-deriving these diagnostics will hit them.

(a) VTHzm is already the eddy flux — do not subtract Vzm·THzm

The long_name, 'Meridional Heat Flux: 3D zon. mean', reads either way. The question was settled against the raw 3-D daily V and T from the h7 tape, forming the true eddy flux directly for six days spanning the seasonal cycle, at four levels and five latitudes (106 comparisons):

Interpretation	Median error vs the true eddy flux	Verdict
VTHzm taken as-is	15% (the h6-interface vs h7-midpoint offset)	CORRECT
VTHzm – Vzm·THzm	1087%	wrong

Taking the subtraction gives a DOWNWARD tropical $\bar{w}^*$ of -0.18 mm/s, which is how the error was caught. The same convention applies to UVzm and UWzm.

(b) Wzm is the log-pressure vertical velocity, not a geometric one

Established by mass continuity. In log-pressure coordinates the divergence of the mean meridional flow must balance the vertical mass-flux divergence; integrating that downward from the model top and regressing the implied w on Wzm gives slope 1.011 with $r = 1.000$, against a predicted 1.112 if Wzm were geometric ($w = -\omega RT/(gp)$, local scale height 6294 m at 215 K). Assuming the wrong one is an 11% error on all upwelling numbers.

(c) The fill sentinel is 1e35 with no _FillValue attribute

xarray does not mask it. Affects 1.34% of cells, all at $p \geq 652$ hPa, so it does not touch the 10 or 70 hPa diagnostics directly - but an unguarded column or global mean is destroyed by it.

(d) MSKtem is not a 0/1 mask

It is a (time, lat, lon) fractional count of above-surface interfaces, taking values 59-71 quantised in 1/48. It has no vertical dimension and cannot be applied to the zonal-mean TEM fields at all.

3 How to read the target bands

The target band is a constraint on the MEAN of a rollout. It is NOT a band that individual years are expected to fall inside. Roughly a third of individual years lie outside a $\pm 0.5\sigma$ band by construction - that is what σ means.

If a model reproduces CESM perfectly, its individual years still scatter with the full interannual standard deviation σ . What must land inside the target band is the average over the whole rollout, whose sampling spread is $\sigma/\sqrt{n}$ and therefore much narrower. For the DJF vortex, $\sigma = 5.69$ m/s but the spread of a 10-year mean is only $5.69/\sqrt{10} = 1.80$ m/s.

This is why the figures in section 4 plot a 10-year running mean on top of the annual values: the running mean is the quantity the band constrains.

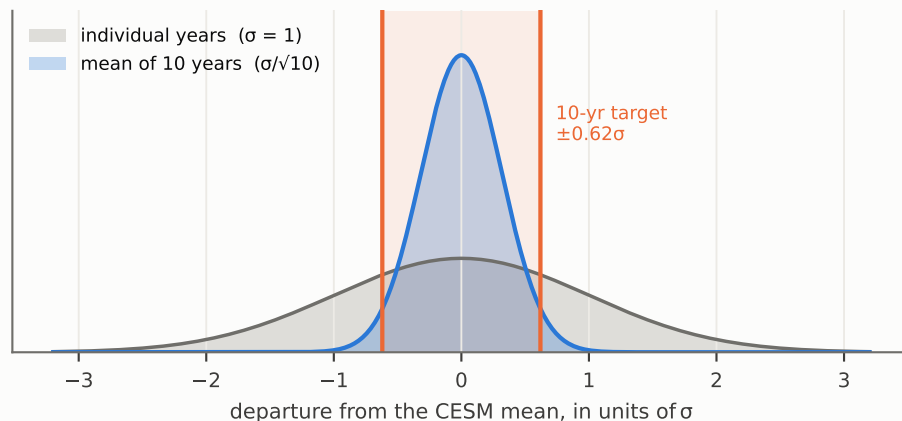
Where the threshold comes from

Two considerations set it, and the operative target is the larger of the two:

- 0.5σ — 'the model error must be smaller than half of natural variability'. This is the standard CCMVal / CCMI framing and does not depend on rollout length.
- $1.96\sigma/\sqrt{n}$ — the detection limit. A threshold tighter than this cannot be distinguished from zero at 95% confidence, so a model sitting exactly on it would be indistinguishable from a perfect one. This does depend on rollout length.

$$\text{target}(n) = \max(0.5 \sigma , 1.96 \sigma / \sqrt{n})$$

The two branches cross at $n = 15.4$ years. At $n = 10$ every bias target is $1.24\times$ its 20-year value.



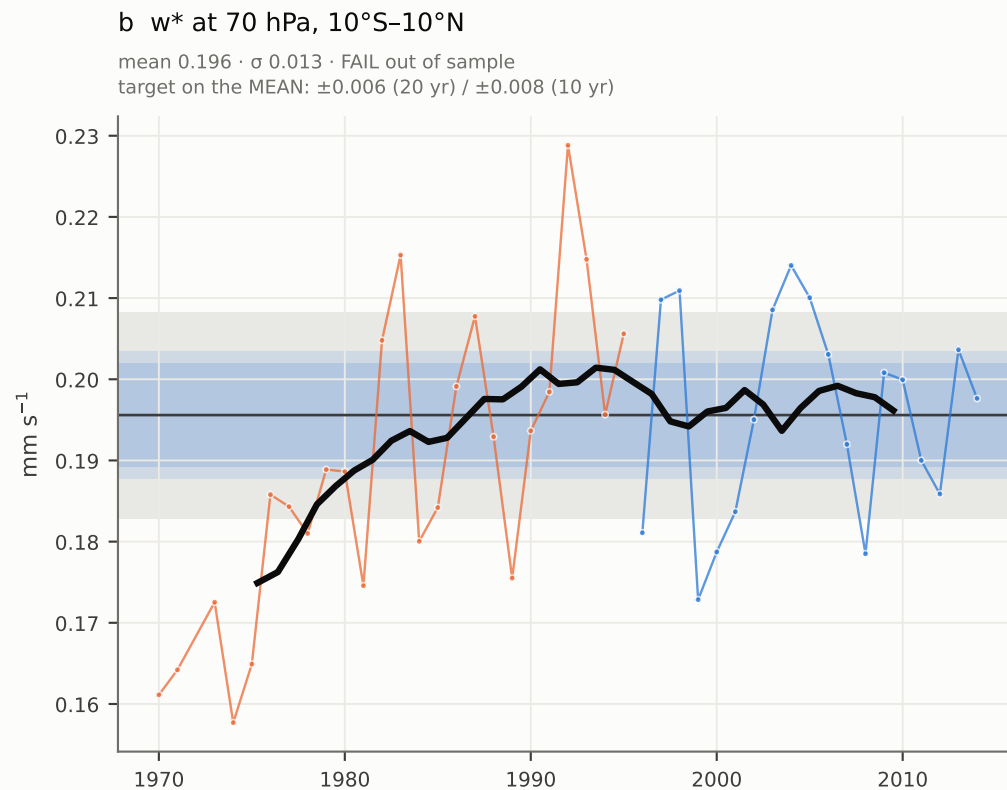
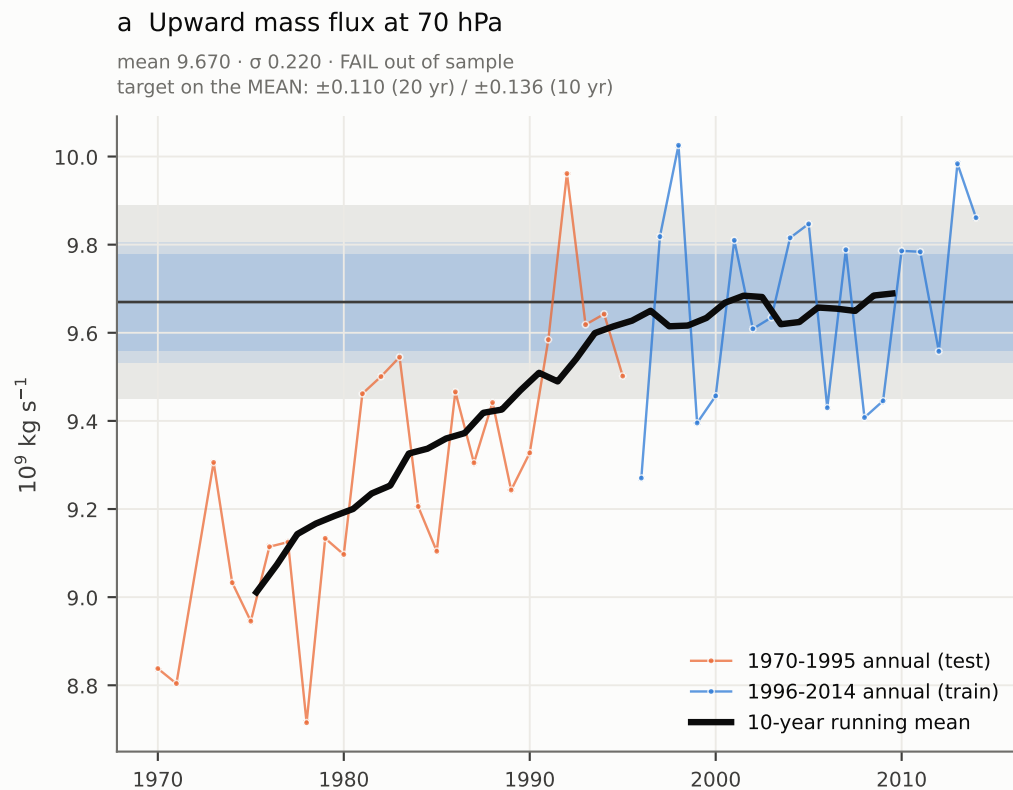
The grey curve is where individual years fall.
The blue curve is where the MEAN of ten of them falls.
The orange band is the target.

A perfect model produces the grey spread and the blue spread simultaneously. Judging it by whether its annual values sit in the orange band would reject it — and would reward a model whose variability was wrongly damped.

This is the same double-penalty effect that makes a plain RMSE target reward under-dispersive models: the variance that minimises MSE is $s = r \cdot \sigma$, which tends to zero as the correlation decays.

4.1 Tropical upwelling — the annual series, 1970-2014

Grey band = $\pm 1\sigma$, where individual years live. Blue bands = the target on the ROLLOUT MEAN (inner 20 yr, outer 10 yr). Black line = the 10-year running mean, which is the quantity the target actually constrains.



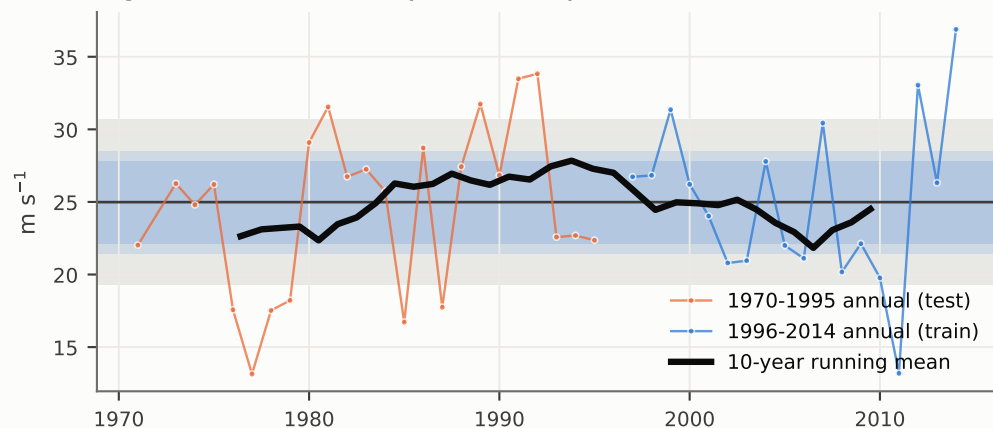
Both upwelling targets FAIL out of sample. The running mean climbs steadily from 9.1 to $9.7 \times 10^9 \text{ kg s}^{-1}$ across the record — a forced Brewer-Dobson acceleration of $+2.1\%/decade$, not internal variability. The 1970–1995 and 1996–2014 means differ by 4.0% , which is $3.5\times$ the target. That is why the upwelling targets must be period-matched rather than absolute (section 7).

4.2 Polar vortex and polar cap temperature — the seasonal series

Same bands as 4.1. All four targets PASS out of sample. Note how far the individual seasons range compared with the running mean: the DJF vortex spans 13 to 37 m s^{-1} while its 10-year mean stays within about $\pm 3 \text{ m s}^{-1}$ — which is exactly why the target applies to the mean and not to individual years.

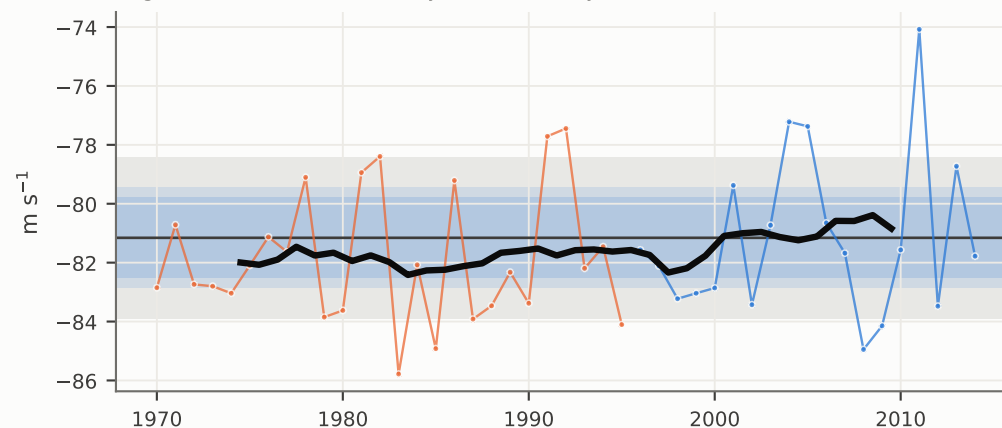
a $\bar{u}$ 60°N, 10 hPa, DJF

mean 24.987 · σ 5.695 · PASS out of sample
target on the MEAN: ± 2.847 (20 yr) / ± 3.530 (10 yr)



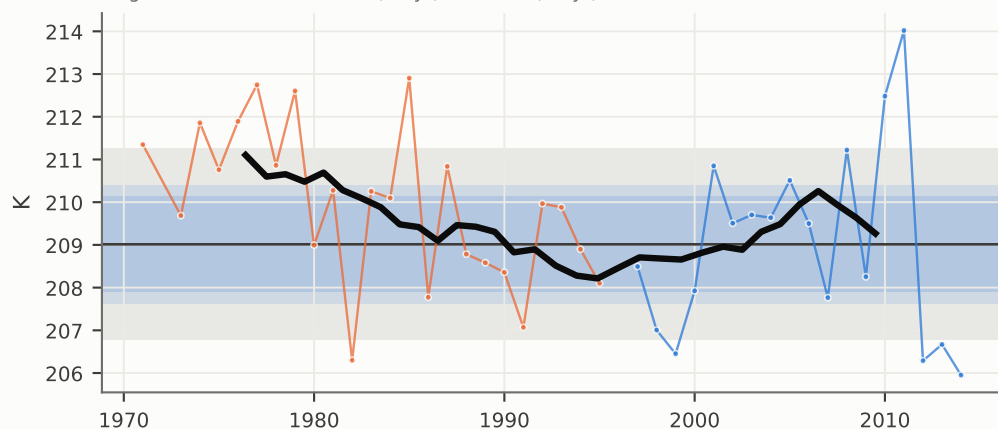
b $\bar{u}$ 60°S, 10 hPa, JJA (sign-flipped)

mean -81.157 · σ 2.749 · PASS out of sample
target on the MEAN: ± 1.374 (20 yr) / ± 1.704 (10 yr)



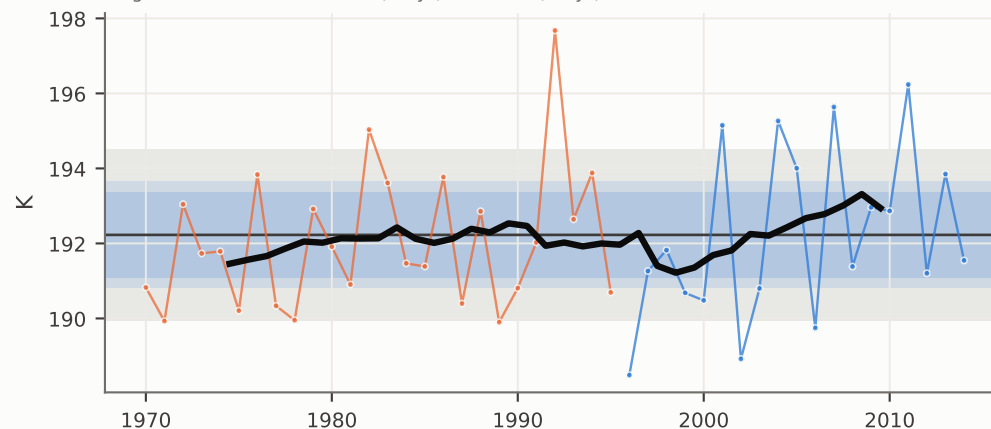
c Polar cap T, 60–90°N, 10–50 hPa, DJF

mean 209.013 · σ 2.242 · PASS out of sample
target on the MEAN: ± 1.121 (20 yr) / ± 1.390 (10 yr)



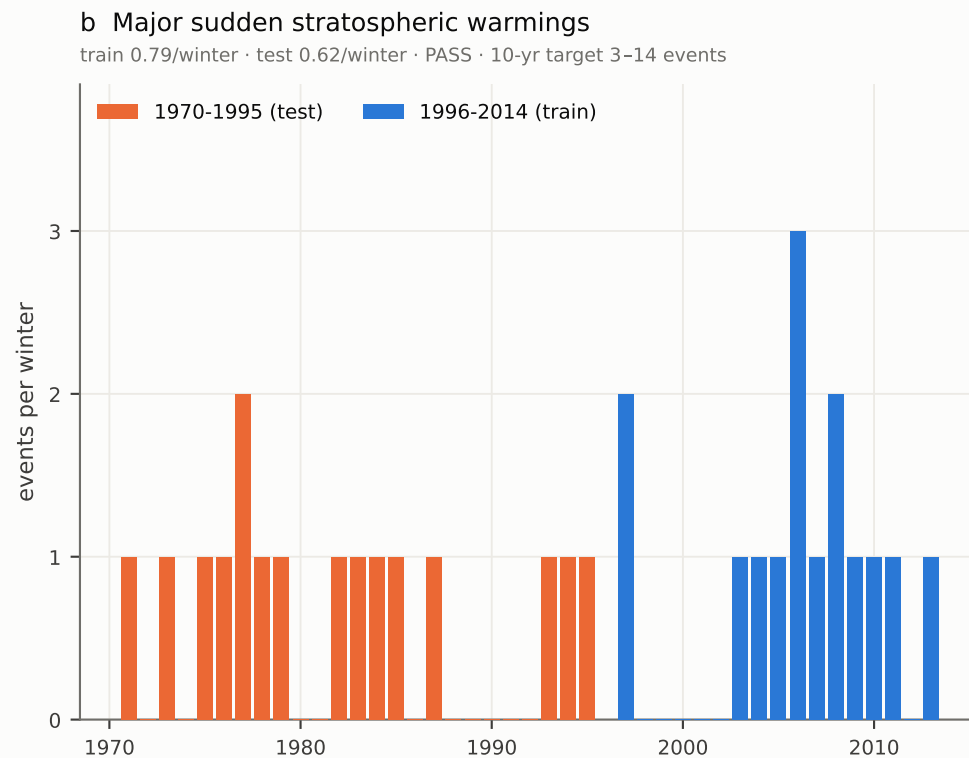
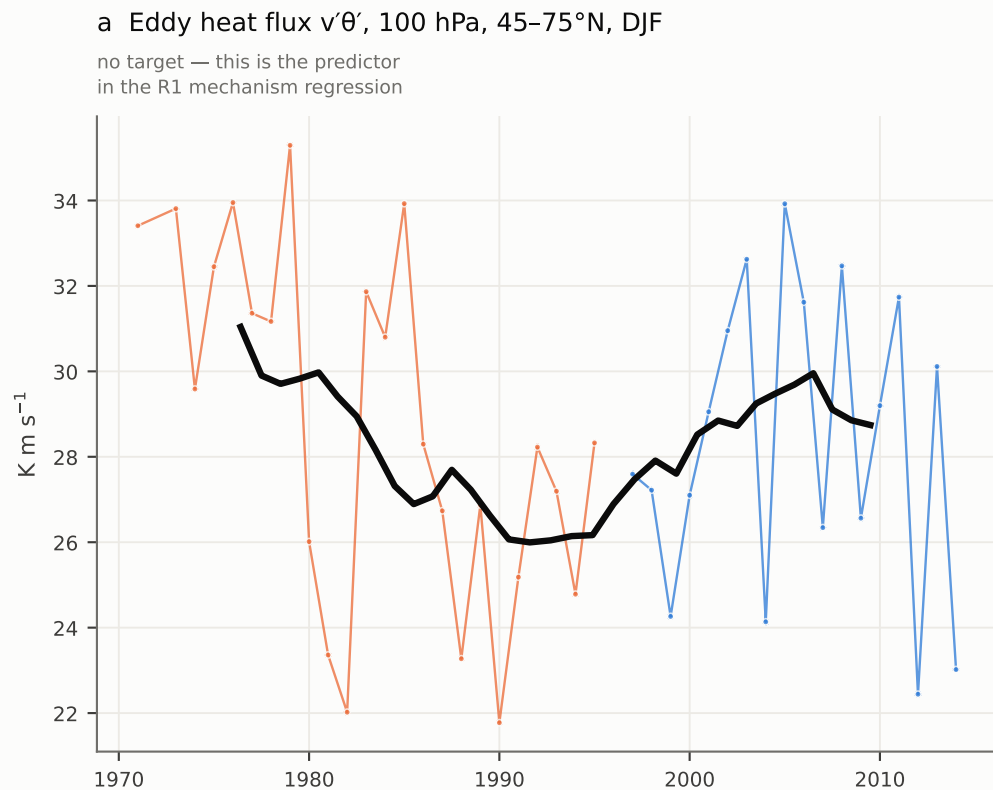
d Polar cap T, 60–90°S, 10–50 hPa, JJA

mean 192.228 · σ 2.287 · PASS out of sample
target on the MEAN: ± 1.144 (20 yr) / ± 1.418 (10 yr)



4.3 Wave driving and sudden stratospheric warmings

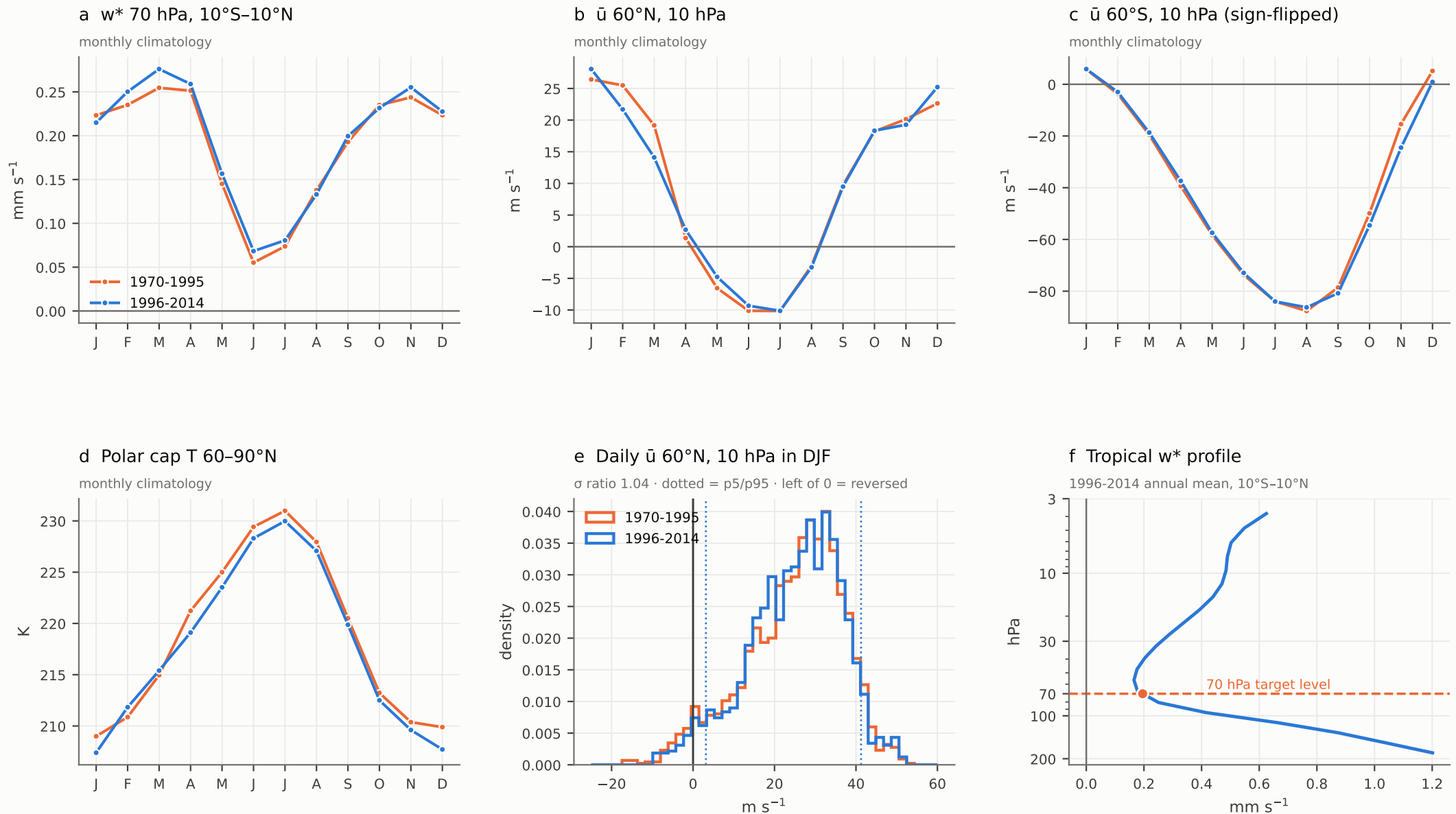
The eddy heat flux at 100 hPa is the upward wave activity entering the stratosphere. It forces both the Brewer-Dobson circulation and vortex breakdown, and is the predictor in the R1 mechanism regression.



31 major warmings in 45 winters, 0.69 per winter overall, close to the observed ~ 0.6 . The Southern Hemisphere produces none in 43 winters, which is also correct — major SH warmings are essentially unobserved. An emulator producing SH midwinter reversals at NH-like rates would be wrong in a way that no mean-state metric would reveal.

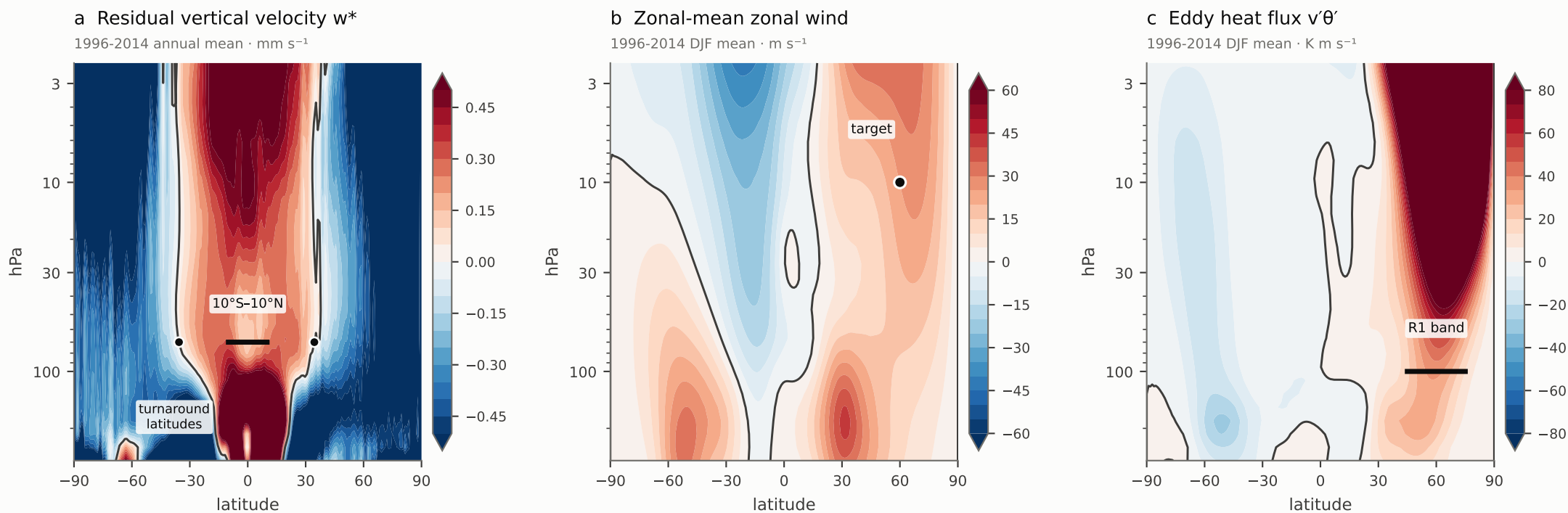
4.4 Seasonal cycles, the daily distribution, and the w^* profile

The two periods' climatologies are nearly identical, so the out-of-sample failures in 4.1 are a shift in the annual MEAN, not a change in seasonal shape.
Tropical w^* peaks in March and is weakest in June.



4.5 Latitude-pressure structure, with the target locations marked

Panel a is the Brewer-Dobson circulation itself: upwelling through the tropics, downwelling at middle and high latitudes, with the zero contour giving the turnaround latitudes that the mass-flux integral runs between.



The mass-flux metric integrates panel a along the 70 hPa line between the two black dots. The vortex metric is the single marked point in panel b. The R1 predictor is the 45–75°N average along the 100 hPa line in panel c.

4.6 Reference values, 1996–2014

Quantity	1996–2014	σ interannual	1970–2014 (45 yr)
Upward mass flux across 70 hPa	$9.670 \times 10^9 \text{ kg s}^{-1}$	0.220 (2.3%)	$9.449 \times 10^9 \text{ kg s}^{-1}$
$\bar{w}^*$ at 70 hPa, 10°S–10°N	0.1956 mm s^{-1}	0.0127 (6.5%)	0.1917 mm s^{-1}
Turnaround latitudes at 70 hPa	35.6°S – 34.5°N	—	—
$\bar{u}$ at 60°N, 10 hPa, DJF	24.99 m s^{-1}	5.69	24.76 m s^{-1}
$\bar{u}$ at 60°S, 10 hPa, JJA	-81.16 m s^{-1}	2.75	-81.57 m s^{-1}
Polar cap T, 60–90°N, 10–50 hPa, DJF	209.01 K	2.24	209.55 K
Polar cap T, 60–90°S, 10–50 hPa, JJA	192.23 K	2.29	192.13 K
NH major SSW	0.789 /winter (15 in 19)	Poisson	0.689 /winter (31 in 45)
SH major SSW	0 in 18 winters	—	0 in 43 winters
Daily DJF σ of $\bar{u}$ 60°N	11.48 m s^{-1}	—	p5/p95 = 3.2 / 41.2
DJF vortex decorrelation time	14 days	—	6.4 independent samples per winter

Forced trends, and why they matter for the anchors

Where a trend is significant, σ is inflated by a forced signal and the detrended value is the honest anchor for internal variability.

Series	Trend per decade	p	σ raw → detrended
70 hPa mass flux	$+0.195 \times 10^9 \text{ kg s}^{-1}$ (+2.1%)	$<10^{-4}$	0.328 → 0.213 (+54% inflation)
$\bar{w}^*$ 70 hPa	$+0.0058 \text{ mm s}^{-1}$ (+3.0%)	0.002	0.0162 → 0.0144
Polar cap T NH DJF	−0.56 K	0.027	2.02 → 1.90
Vortex NH DJF	$+0.60 \text{ m s}^{-1}$	0.41 (ns)	no correction
Vortex SH JJA	$+0.35 \text{ m s}^{-1}$	0.22 (ns)	no correction

The Brewer-Dobson acceleration of +2.1%/decade matches the ~2%/decade reported across chemistry-climate models. Vortex strength has no significant trend in either hemisphere, so those anchors need no correction — which is part of why the vortex targets transfer out of sample and the upwelling targets do not.

5 The target table

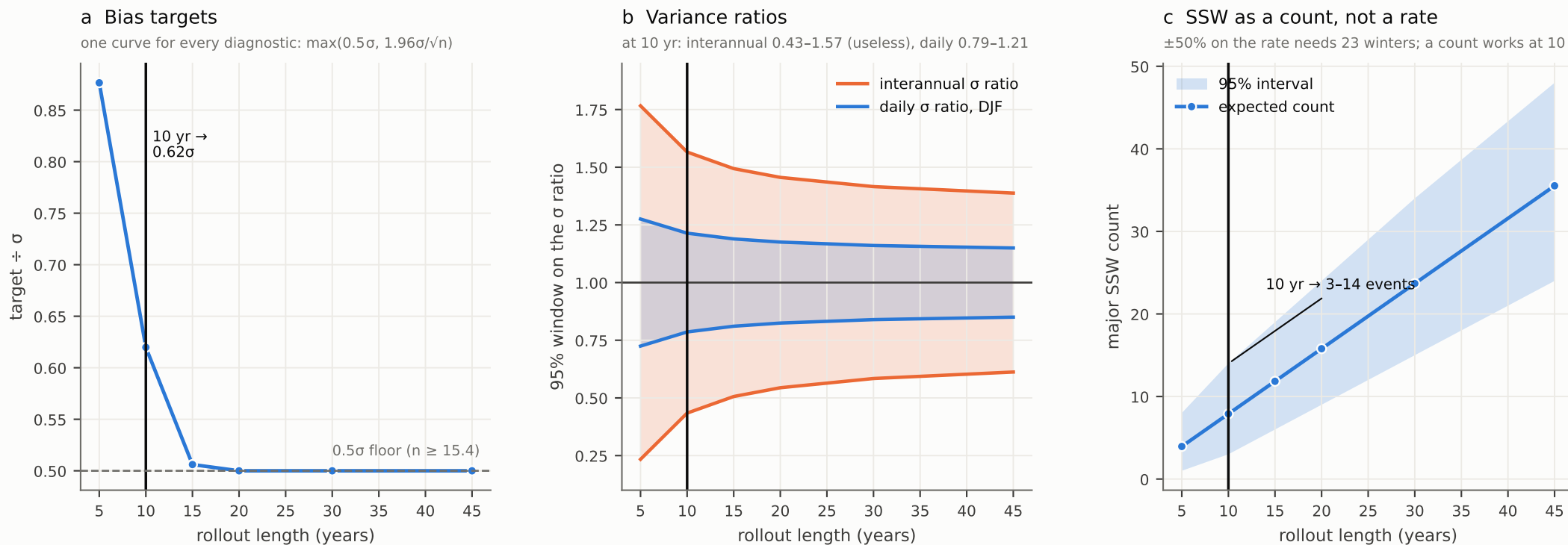
Thresholds are on the ROLLOUT MEAN (see section 3), except the variance ratios, counts and slopes, which are on the quantity named.
Full notes in output/validation_targets.csv.

Diagnostic	10-year target	20-year target	Out-of-sample
Upward mass flux across 70 hPa, between the turnaround latitudes [PRIMARY]	0.136 (1.4%)	0.110 (1.1%)	FAIL (3.54x target)
w* at 70 hPa, cos-weighted 10S-10N [SECONDARY]	0.0079 (4.0%)	0.0064 (3.3%)	FAIL (1.07x target)
Interannual sigma ratio of the 70 hPa mass flux	0.43 - 1.57 (too wide to use)	0.54 - 1.46	PASS (ratio 0.87)
Forced BDC trend, 70 hPa mass flux [EXPECTATION, not a gate]	not testable	not testable (needs ~30 yr)	-
u at 60N, 10 hPa, DJF mean [PRIMARY]	3.5 m/s	2.8 m/s	PASS (0.14x target)
u at 60S, 10 hPa, JJA mean [strongest drift detector]	1.7 m/s	1.4 m/s	PASS (0.53x target)
Daily sigma ratio of u at 60N, 10 hPa in DJF	0.79 - 1.21	0.82 - 1.18	PASS (ratio 1.04)
Major SSW frequency (Charlton-Polvani 2007)	3-14 events (expect 7.9)	9-24 events (expect 15.8)	PASS (0.62/winter)
5th / 95th percentile of daily u at 60N, 10 hPa in DJF	5 m/s on each	5 m/s on each	PASS (1.0 / 41.4)
Major SSW frequency, SH	0 events	<= 1 event	PASS (0 in both periods)
T, 60-90N, 10-50 hPa, DJF mean	1.4 K	1.1 K	PASS (0.84x target)
T, 60-90S, 10-50 hPa, JJA mean	1.4 K	1.1 K	PASS (0.15x target)
R1 u(60N,10hPa) vs DJF v'theta' at 100 hPa, 45-75N	CI must OVERLAP the CESM CI (rollout CI ~ +/-0.64)	CI must OVERLAP the CESM CI	FAIL (test slope -0.714 [-1.132,-0.336])
R2 u(60N,10hPa) vs DJF polar cap T, 60-90N, 10-50 hPa	CI must OVERLAP the CESM CI (rollout CI ~ +/-0.85)	CI must OVERLAP the CESM CI	PASS (test slope -2.111 [-3.054,-1.211])

Rows in orange fail their own out-of-sample check: CESM 1970–1995 does not meet a target set on CESM 1996–2014. They are not usable as absolute thresholds — see section 7.

6.1 How each target scales with rollout length

Bias targets widen smoothly and stop improving beyond 15.4 years. The variance and event metrics change character rather than merely widening: the interannual variance ratio becomes untestable, and SSW is better stated as a count.



6.2 What a 10-year rollout can and cannot test

Can be tested at 10 years

- All six bias targets, at 1.24× their 20-year thresholds: mass flux 1.4%, w^* 4.0%, $\bar{u}$ 60°N 3.5 m/s, $\bar{u}$ 60°S 1.7 m/s, polar cap T 1.4 K in each hemisphere.
- NH major SSW count: 3–14 events, against an expectation of 7.9.
- SH major SSW count: 0 events. A binary falsification test.
- Daily σ ratio of the DJF vortex: 0.79–1.21. The only variance metric that survives at this length, because it has 6.4 independent samples per winter rather than one.
- 5th and 95th percentiles of the daily DJF vortex, to 5 m/s each.
- Both mechanism slopes: R1 to ± 0.64 , R2 to ± 0.85 .

Cannot be tested at 10 years

- The interannual σ ratio. Its 95% window is 0.43–1.57, so nothing realistic fails it and reporting a pass is meaningless. Use the daily σ ratio instead.
- The forced BDC trend. It needs roughly 30 years to separate from noise — and AIDE-WACCM cannot reproduce it in any case, because GHG concentrations are not among its input-only forcings (TOA solar, time-of-day, year-progress, statics).
- Any absolute upwelling bias measured against a different CESM epoch.

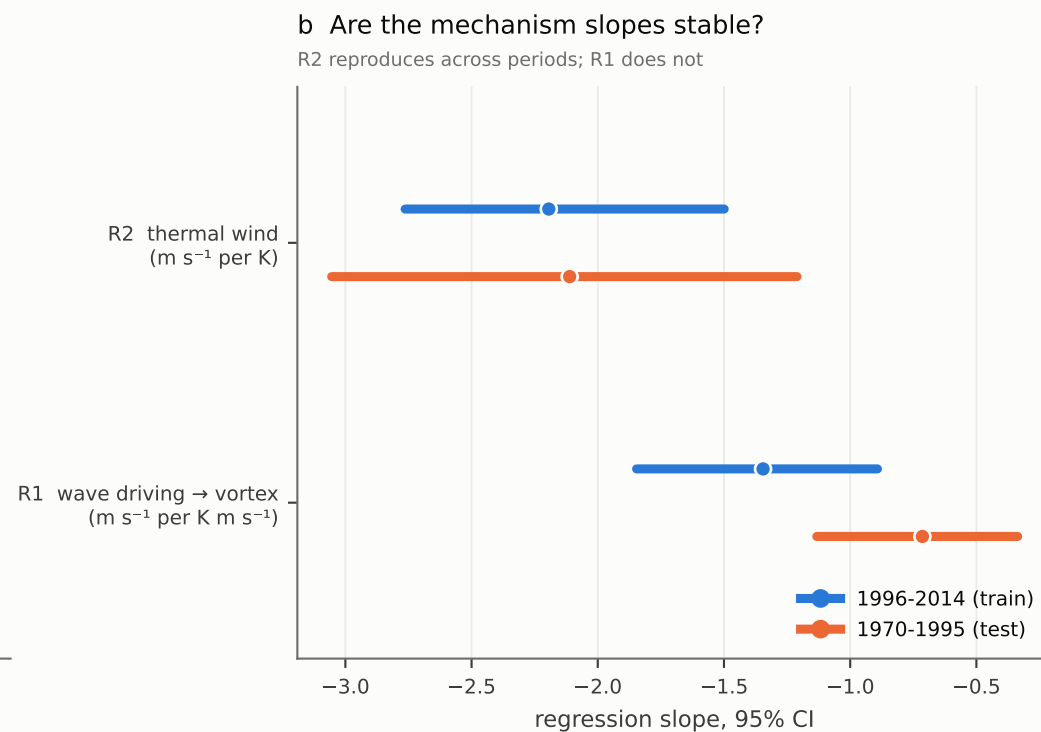
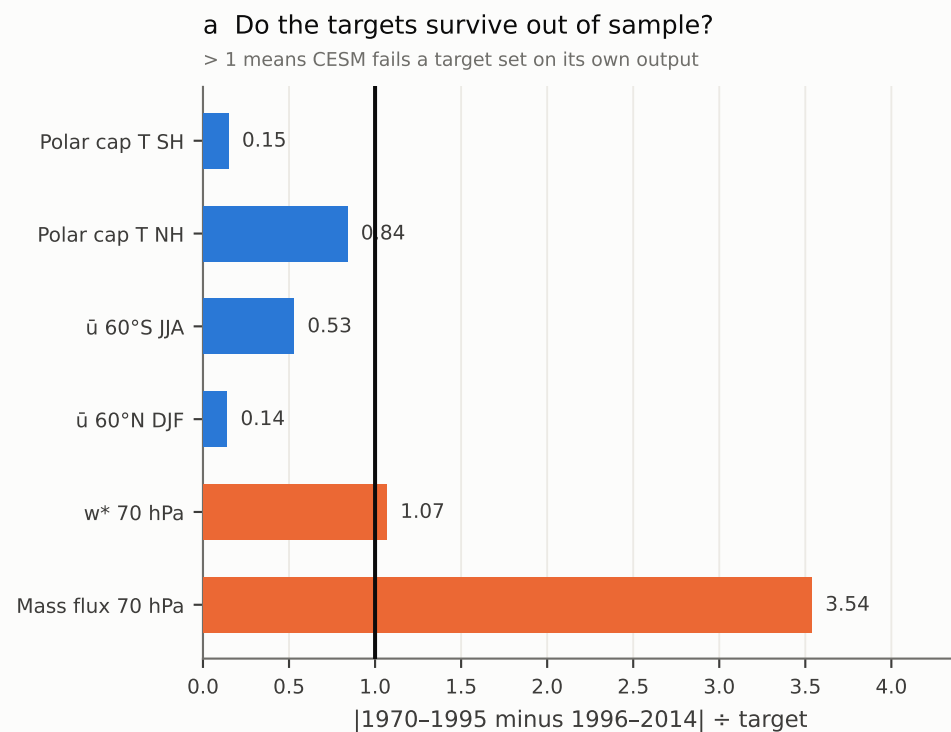
Two reframings matter more than the widened numbers. SSW should be reported as a COUNT with a Poisson interval, not a rate $\pm 50\%$ — the rate framing needs 23 winters before $\pm 50\%$ is resolvable at all, whereas a count interval is well defined at any length. And the mechanism slopes become relatively MORE valuable at short rollout length, since they are resolvable in 3–6 winters while the variance metrics degrade fastest.

Why the reference σ should still come from the full record

σ has two distinct jobs. As a statement about nature it should be estimated from the longest record available — a 10-year rollout does not change what CESM's natural variability is. As a limit on detection it enters as $\sigma/\sqrt{n}$, which is where the rollout length belongs. Shortening the record used to estimate σ would only add noise to the anchor; shortening the rollout is what widens the threshold.

7.1 Out-of-sample verification: targets set on 1996-2014, scored on 1970-1995

10 of 13 targets pass. A target that CESM itself cannot meet on a different sample of its own output is not a target — it is an artefact of the period it was fitted to.



7.2 Reading the out-of-sample result

The upwelling targets fail, and the reason is a forced trend

The 1970–1995 mass flux is 4.0% lower than 1996–2014 — 3.5× the target. That is not sampling noise. The trend is $+0.31 \times 10^9 \text{ kg s}^{-1}$ per decade within 1970–1995 ($p < 10^{-4}$) and statistically absent within 1996–2014 ($p = 0.34$), so the Brewer-Dobson acceleration is concentrated in the earlier period and the two epochs have genuinely different means.

The consequence is a protocol requirement, not a looser number. The upwelling targets are PERIOD-MATCHED, not absolute: compare a rollout against the CESM years it actually covers, or detrend both sides. Applying a 1.4% bias target across epochs would fail a perfect emulator.

This connects to open question 7.1(10) in the state document. AIDE-WACCM's input-only forcings are TOA solar, time-of-day, year-progress and statics, so a free rollout has no mechanism by which to reproduce a GHG-forced trend in the first place. Until GHG concentrations become an input, the trend is a documented expectation rather than a gate.

R1 is not stable across periods; R2 is

The wave-driving slope is -1.35 [$-1.85, -0.89$] on 1996–2014 but -0.71 [$-1.13, -0.34$] on 1970–1995 — outside the training CI, though the two CIs do overlap. The test must therefore be CI OVERLAP rather than containment, and R1 is not the primary mechanism test.

Thermal wind is the reliable one: -2.19 [$-2.76, -1.50$] against -2.11 [$-3.05, -1.21$], and resolvable in 3 winters. On the pooled record the slope is -2.06 m s^{-1} per K.

v2's documented -2 to -3 K cold pole should therefore come with a $+4$ to $+6 \text{ m s}^{-1}$ vortex bias if the model is thermally consistent. If v2 has the cold pole but not the vortex bias, that is a structural error rather than a tuning error — and open question 7.1(5), 'whose cold bias?', can be closed by checking that one pair of numbers.

What passes

All four vortex and polar-cap bias targets pass at 0.14–0.84× their threshold, as do the NH and SH SSW counts, the daily σ ratio (1.04) and the DJF percentiles. Those transfer across epochs and can be stated absolutely.

8 Two protocol requirements that apply at any rollout length

(a) Pin the estimator, and reduce CESM truth through the identical code path

The two standard routes to $\bar{w}^*$ — the direct Andrews-Holton-Leovy form used here, and integrating the residual meridional velocity $\bar{v}^*$ through continuity — differ by 10.8% on identical data (0.1956 against 0.2168 mm s⁻¹ for the tropical mean at 70 hPa). A 1.4% bias target is therefore not a statement about the physical Brewer-Dobson circulation; it is a statement about a specific estimator, and is only valid as a model-minus-truth comparison in which both sides went through the same reduction. That reduction is `aide_val_common.py:tem_residual`.

(b) Score on the emulator's own grid, against truth reduced onto that grid

An emulator outputs 6-hourly snapshots on 20 or 45 coarse hybrid layers and no eddy fluxes, so $\bar{w}^*$ must be rebuilt from snapshots. Recomputing the diagnostics from the preprocessed training stores — which are CESM truth, coarsened and subsampled exactly as the emulator sees it — against the native 70-level answer for year 2000:

Dagnostic	v5 (20 levels)	v7 (45 levels)	Target size	Error ÷ target, v5 / v7
$\bar{w}^*$ 70 hPa	+2.0%	+11.8%	3.8%	0.54 / 3.13
Mass flux 70 hPa	+1.4%	+3.3%	1.1%	1.26 / 2.95
$\bar{u}$ 60°N 10 hPa DJF	−0.6%	−1.4%	11.3%	0.05 / 0.12
Polar cap T DJF	−0.88 K	+0.20 K	0.95 K	0.93 / 0.21
Daily σ , DJF	+5.3%	+2.4%	—	—

The vortex targets survive coarsening with an order of magnitude to spare and can be stated absolutely. The upwelling targets cannot: the estimator error is comparable to or larger than the target on both grids. Note that the FINER v7 grid is WORSE — $\bar{w}^*$ is a small residual of two larger terms and is sensitive to the $\partial\theta/\partial z$ stencil, so this is not evidence that v7 resolves less. It is an argument for same-grid scoring, where the errors largely cancel. `/data/v6/rollout/zonal_reduce_truth.py` already implements that pattern.

This measures the error in DIAGNOSING the circulation on a coarse grid from truth. It does not show that coarse layers are harmless for SIMULATING it, which is a separate and still-open question.

9 What changed, and what this does not cover

Relative to §6.2 of claude-state-11August2026.md

Proposed there	Here	Why
$\bar{w}^*$ bias $\leq 3\%$ of the mean	1.4% on mass flux at 10 yr, and period-matched rather than absolute	3% sat below the 10.8% estimator spread; and an absolute cross-epoch target fails on CESM's own data
Interannual σ ratio 0.9–1.1	unusable at 10 yr; use the daily σ ratio 0.79–1.21	$\pm 10\%$ is far inside the sampling noise of a 10- or 20-year variance estimate
Vortex DJF bias $\leq 3 \text{ m s}^{-1}$	$\leq 3.5 \text{ m s}^{-1}$ at 10 yr, 2.8 at 20 yr	the guess was right at 20 yr — 0.5σ measures 2.85
Daily σ ratio 0.9–1.1	0.79–1.21 at 10 yr	$\tau = 14 \text{ d}$ gives only 6.4 independent samples per winter
SSW within $\pm 50\%$	a count: 3–14 events in 10 winters	$\pm 50\%$ on the rate needs 23 winters; a count works at any length
Polar cap T bias $\leq 1\text{--}2 \text{ K}$	$\leq 1.4 \text{ K}$ at 10 yr, 1.1 at 20 yr	$0.5\sigma = 1.12 \text{ K}$
—	SH vortex $\leq 1.7 \text{ m s}^{-1}$, SH SSW = 0	new; $3.3\times$ stronger and $2.1\times$ less variable, so the sharpest drift detector available
wave driving $\rightarrow$ vortex, build first	quantified and demoted; require CI overlap	R1 fails its own out-of-sample check
upwelling $\leftrightarrow$ AoA consistency	thermal wind quantified instead	upwelling $\rightarrow$ polar T has $r = -0.09$ and would need 509 seasons

Limitations

- Fixed SST: no ENSO, so the interannual σ anchoring most targets is smaller than the real atmosphere's. Correct for scoring against this CESM run; wrong for claiming realism.
- The two segments are separate CESM runs joined at a restart. Part of the 4% mass-flux step could be a restart discontinuity rather than a trend; the significant within-segment trend argues against that, but they have not been separated.
- The observability test used one year (2000) — enough to size the estimator error, not to characterise its year-to-year spread. Worth repeating on 3–5 years before section 8's numbers are treated as fixed.
- Age of Air is not covered. AOA1/AOA2 exist in the h0 monthly stream, but converting the tracer mixing ratio to an age in years needs the tracer's boundary specification, which is not on disk.
- SSW detection is my implementation of Charlton-Polvani, not a community-shared one. The rate is consistent with the literature but the central dates have not been cross-checked against an independent catalogue.
- Nothing here has been run against an actual emulator rollout — every number is CESM against CESM.

10 Appendix: time handling in the training-data preprocessing

Finding the h1↔h6 index offset in section 1.4 raised an obvious question: does `/data/preprocess_cesm.py`, which builds the emulator training store, have the same problem? It does not — but the check turned up two related issues that do affect the training data.

Clean: the index offset is not present

`process_year` selects the 6-hourly subset by DECODED TIMESTAMP (`t.hour % 6 == 0`), and `load_ps_6hourly` / `load_ts_6hourly` interpolate against decoded timestamps rather than record indices. The offset that bites index arithmetic therefore never arises. `preprocess_cesm_v5.py` and `_v7.py` share the same structure.

Issue 1: every averaged field is placed half an averaging window late

No script reads `time_bnds`. CAM stamps averaged output at the END of the interval — verified on h1, h6 and h0, where the time coordinate equals the upper bound exactly. Interpolating 'to the timestamp' therefore treats a mean over $[t_0, t_1]$ as if it were valid at t_1 rather than at the midpoint, shifting the series late by half the window. Measured on 1980:

Field	Tape	Lag	RMS error	p99	Max	Consequence
PS	h6 daily	12 h	2.66 hPa	9.4 hPa	28.1 hPa	displaces tropospheric hybrid levels by up to 28 hPa; the stratosphere is untouched because <code>hybm = 0</code> above ~182 hPa
TS	h0 monthly	15 d	2.08 K	6.6 K	15.0 K	the <code>surface_temperature</code> channel of the training store is ~15 days out of phase; largest over land in spring and autumn
N2O	h0 monthly	15 d	—	—	—	same shift, on the <code>nitrous_oxide</code> channels
3-D fields	h1 hourly	30 min	—	—	—	negligible

Issue 2: `_monthly_mean_ps` is paired one month late

It groups daily PS by the (year, month) of the MONTHLY timestamp. Because that stamp is the interval end, the January N2O field — stamped 1 February — is converted from hybrid to pressure using FEBRUARY's mean surface pressure, and so on through the record. The effect is small, since monthly-mean PS changes little month to month, but it is wrong.

Neither issue affects any number in this report: the validation diagnostics are read from h6 directly, and both target levels sit above 182 hPa where `hybm = 0` and surface pressure is irrelevant. They affect the TRAINING STORE. The fix is one line in each loader — interpolate against `0.5*(time_bnds[:,0] + time_bnds[:,1])` instead of `time` — and regenerating is the owner's call, not mine.

Caveat on framing: linear interpolation between monthly means does not preserve the monthly mean anyway, so 'correct' here is the conventional midpoint choice, not an exact reconstruction. Measured by `scripts/11_check_preprocess_time.py`.